1RV24MC042_HARSHAVARDHAN_BR_LAB10

Demonstrate Kubernetes Dashboard and CLI for Cluster Monitoring

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Project Structure

```
❯ tree
.
├── deployments.yml
└── services.yml

1 directory, 2 files
```

Step 1 : deployments.yaml

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
  labels:
    app: nginx
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
        - name: nginx
          image: nginx:latest
          ports:
            - containerPort: 80
```

Step 2 : services.yaml

```
❯ cat services.yml
apiVersion: v1
kind: Service
metadata:
  name: nginx-service
spec:
  selector:
    app: nginx
  ports:
    - protocol: TCP
      port: 80
      targetPort: 80
  type: NodePort
```

Step 3: Apply Deployment

```
kubectl apply -f deployment.yaml
```

Step 4: Get pods

```
kubectl get pods
```

Step 5: Apply Service

```
kubectl apply -f service.yaml
```

Step 6 : Get services

```
kubectl get svc
```

```
❯ ls
deployments.yml  services.yml
❯ kubectl apply -f deployments.yml
deployment.apps/nginx-deployment created
❯ kubectl get pods
No resources found in default namespace.
❯ kubectl apply -f service.yaml
error: the path "service.yaml" does not exist
❯ kubectl apply -f services.yml
service/nginx-service created
❯ kubectl get svc
NAME               TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)        AGE
kubernetes         ClusterIP   10.96.0.1       <none>        443/TCP        7d2h
nginx-deployment   NodePort    10.98.31.82     <none>        80:30214/TCP   6d23h
nginx-service      NodePort    10.100.210.22   <none>        80:30426/TCP   6s

 ~/Documents/Obsidian Vault/RVCE_/SEM3/DEVOPS/devopslab/p9_10/p10  ✔  10:10:32 AM
```

Step 7: Monitor Using CLI

```
kubectl get nodes
```

Step 8 : Describe a Pod

```
kubectl describe pod <pod-name>
```

```
❯ kubectl get pods
No resources found in default namespace.
❯ kubectl get pods --all-namespaces
```

Step 9: View Pod Logs

```
kubectl logs pod1
```

```
❯ kubectl get deployment
NAME               READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment   0/2     0            0           3m32s
❯ kubectl describe deployment nginx-deployment
Name:                   nginx-deployment
Namespace:              default
CreationTimestamp:      Fri, 19 Dec 2025 10:09:53 +0530
Labels:                 app=nginx
Annotations:            deployment.kubernetes.io/revision: 1
Selector:               app=nginx
Replicas:               2 desired | 0 updated | 0 total | 0 available | 2 unavailable
StrategyType:           RollingUpdate
MinReadySeconds:        0
RollingUpdateStrategy:  25% max unavailable, 25% max surge
Pod Template:
  Labels:  app=nginx
  Containers:
   nginx:
    Image:         nginx:latest
    Port:          80/TCP
    Host Port:     0/TCP
    Environment:   <none>
    Mounts:        <none>
  Volumes:         <none>
  Node-Selectors:  <none>
  Tolerations:     <none>
Conditions:
  Type             Status  Reason
  ----             ------  ------
  Progressing      True    NewReplicaSetCreated
  Available        False   MinimumReplicasUnavailable
  ReplicaFailure   True    FailedCreate
OldReplicaSets:    <none>
NewReplicaSet:     nginx-deployment-6f9664446b (0/2 replicas created)
Events:
  Type    Reason             Age    From                   Message
  ----    ------             ----   ----                   -------
  Normal  ScalingReplicaSet  3m46s  deployment-controller  Scaled up replica set nginx-deployment-6f9664446b from 0 to
 2
```

Step 10: Dashboard

```
minikube dashboard
```